MINI PROJECT SET 01

Student Academic Performance Analysis and Prediction

Problem statement: A college wants to understand the factors that influence students' academic performance and predict whether a student is likely to pass or require academic support.

Suggested dataset: Use a student-performance dataset (for example, UCI Student Performance) or create a documented synthetic dataset with attendance, study hours, assignment scores, previous marks, and final result.

Tasks to be completed

1. Load the dataset using Pandas and inspect its shape, columns, data types, summary statistics, and class distribution.
2. Clean the data by handling missing values, duplicates, inconsistent entries, and unsuitable columns. Encode categorical features and scale numerical features where appropriate.
3. Use regression to predict a continuous outcome such as final marks. Plot actual versus predicted values and report suitable regression metrics.
4. Classify pass/fail using KNN and Naïve Bayes. Print correct and incorrect predictions.
5. Train SVM classifiers using Linear, Polynomial, and RBF kernels; compare their test accuracy.
6. Train an ID3-style decision tree using entropy/information gain (or a documented entropy-based decision tree implementation) and a Random Forest. Vary the number of trees and compare performance.
7. Apply K-Means to selected scaled features; test at least three values of k and interpret the clusters.
8. Visualize important patterns and summarize practical interventions suggested by the results.

Evaluation measures: Regression: MAE, MSE/RMSE, R^2. Classification: accuracy, precision, recall, F1-score, confusion matrix. Clustering: inertia/elbow plot and cluster interpretation.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 02

Breast Cancer Classification and Risk-Feature Analysis

Problem statement: Build and compare machine-learning models that classify a breast tumour as benign or malignant using measured cell-nucleus features. This is an educational dataset exercise, not a clinical diagnostic tool.

Suggested dataset: Use scikit-learn's Wisconsin Diagnostic Breast Cancer dataset (load_breast_cancer) or an equivalent public dataset.

Tasks to be completed

1. Load the dataset, document the feature names and target labels, and inspect class balance and feature statistics.
2. Check for missing values and duplicates; split data into training and testing sets using stratification. Scale features for distance- and margin-based models.
3. Fit a regression model to predict one selected continuous tumour measurement from other features and plot actual versus predicted values.
4. Train KNN and Gaussian Naïve Bayes classifiers. Display correct and incorrect test predictions with actual and predicted labels.
5. Train SVM models with Linear, Polynomial, and RBF kernels and compare their metrics.
6. Implement an ID3-style entropy-based decision tree or use a documented entropy criterion as an ID3-style approximation; compare it with Random Forest.
7. Vary Random Forest n_estimators (for example, 10, 50, 100, and 200) and record performance.
8. Apply K-Means to scaled features for k = 2, 3, and 4. Compare clusters with the known labels for learning purposes and discuss limitations.

Evaluation measures: Accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC if covered; regression MAE/RMSE/R^2; K-Means inertia and cluster interpretation.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 03

House Price Estimation and Neighbourhood Segmentation

Problem statement: A real-estate analyst needs to estimate house prices from property characteristics and group properties into similar segments.

Suggested dataset: Use a public housing dataset such as California Housing for regression, plus a compatible property dataset with a categorical price band for classification; alternatively, create a documented dataset with property area, bedrooms, age, location category, and price.

Tasks to be completed

1. Load and explore the housing data using Pandas; identify missing values, duplicates, outliers, and numerical/categorical columns.
2. Preprocess the data, encode categorical columns, and scale the features used by KNN, SVM, and K-Means.
3. Train a regression model to predict house price and plot actual versus predicted prices and residuals.
4. Create price categories (such as low, medium, and high) using documented thresholds, then classify them with KNN and Naïve Bayes.
5. Compare SVM Linear, Polynomial, and RBF kernels for price-category classification.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary the number of trees and compare results.
7. Use K-Means for k = 2, 3, 4, and 5 to group properties. Plot clusters using two meaningful features and explain each segment.
8. Discuss which property characteristics appear useful and the limitations of using the model for real market decisions.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score and confusion matrix; K-Means elbow plot.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 04

Used-Car Price Prediction and Vehicle Segmentation

Problem statement: A used-car marketplace wants to estimate vehicle prices and classify listings into price bands to support listing analysis.

Suggested dataset: Use a public used-car dataset or a clearly documented dataset containing year, kilometres driven, fuel type, transmission, owner count, engine capacity, and selling price.

Tasks to be completed

1. Load and inspect the dataset; identify data quality issues, missing values, duplicate listings, and unusual price or mileage values.
2. Clean and transform the data, encode fuel/transmission/owner categories, and scale appropriate features.
3. Use regression to predict selling price and plot predicted versus actual values.
4. Create price-band labels and train KNN and Naïve Bayes classifiers; show examples of correct and incorrect predictions.
5. Compare SVM classifiers with Linear, Polynomial, and RBF kernels.
6. Build an entropy-based ID3-style decision tree and Random Forest. Evaluate multiple values of n_estimators.
7. Apply K-Means to scaled vehicle features for k = 2, 3, 4, and 5; describe the resulting vehicle segments.
8. Summarize the most useful factors for price estimation and the limitations of the dataset.

Evaluation measures: MAE/RMSE/R^2; accuracy, precision, recall, F1-score, confusion matrix; inertia/elbow plot.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 05

Telecom Customer Churn Prediction

Problem statement: A telecom provider wants to identify customers who may discontinue service and explore customer groups with different usage or billing patterns.

Suggested dataset: Use the IBM Telco Customer Churn public dataset or a similar dataset with tenure, services, monthly charges, contract type, payment method, and churn label.

Tasks to be completed

1. Load the dataset and explore churn rate, feature types, missing values, and possible data-quality issues.
2. Clean and preprocess the data, convert total charges or other numeric fields correctly, encode categorical variables, and scale numerical features where required.
3. Fit a regression model to estimate monthly charges or tenure from other attributes and visualize actual versus predicted values.
4. Train KNN and Naïve Bayes models to predict churn. Print correct and incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels for churn classification.
6. Train an entropy-based ID3-style decision tree and Random Forest. Vary the number of trees and compare the results.
7. Use K-Means with k = 2, 3, 4, and 5 on selected scaled customer attributes; profile the clusters.
8. Explain how the findings could help customer-retention planning without claiming that correlation proves causation.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; cluster profiles and elbow plot.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 06

Loan Approval Prediction and Applicant Segmentation

Problem statement: A lending organisation wants to study applicant data, classify historical loan approval outcomes, and group applicants by financial characteristics. The model must be treated as an educational exercise and not used for actual lending decisions.

Suggested dataset: Use a public loan-approval dataset with income, loan amount, credit history, employment, property area, and approval outcome. Document the dataset source and its limitations.

Tasks to be completed

1. Load and inspect the data, target balance, missing values, and categorical/numerical features.
2. Handle missing values, encode categorical variables, and scale suitable features. Avoid data leakage by fitting transformations only on training data.
3. Use regression to predict a continuous field such as loan amount from applicant attributes and plot actual versus predicted values.
4. Train KNN and Naïve Bayes classifiers for the historical approval label and print correct and incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels.
6. Train an entropy-based ID3-style decision tree and Random Forest; compare performance while varying the number of trees.
7. Apply K-Means to selected scaled financial attributes for k = 2, 3, and 4; describe clusters without assigning moral or creditworthiness labels.
8. Discuss fairness, privacy, class imbalance, and why the result must not be treated as an automatic real-world approval system.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 07

Retail Sales Forecasting and Store/Item Grouping

Problem statement: A retail business wants to estimate sales from product and store information, classify sales levels, and discover groups of similar items or stores.

Suggested dataset: Use a public retail sales dataset or a documented dataset with item category, store type, price, promotions, date-related features, and sales quantity/revenue.

Tasks to be completed

1. Load and inspect the data; parse dates where applicable and check missing values, duplicates, and inconsistent product/store labels.
2. Create useful features, encode categorical columns, and scale features for distance-based methods.
3. Use regression to estimate sales quantity or revenue. Use a time-aware split if observations are chronological; plot actual versus predicted sales.
4. Define low/medium/high sales classes using stated thresholds, then train KNN and Naïve Bayes classifiers.
5. Compare SVM Linear, Polynomial, and RBF kernels for sales-level classification.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary the number of trees and compare performance.
7. Use K-Means for k = 2, 3, 4, and 5 to group products or stores and interpret the resulting groups.
8. Explain how the analysis could support inventory planning and state limitations.

Evaluation measures: MAE/RMSE/R^2; accuracy, precision, recall, F1-score; cluster profiles and elbow plot.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 08

Diabetes Dataset Classification and Feature Exploration

Problem statement: Compare machine-learning classifiers on a public dataset used to study diabetes-related outcome labels. This is an educational exercise and must not be used for diagnosis or treatment decisions.

Suggested dataset: Use a public diabetes classification dataset such as the Pima Indians Diabetes dataset, respecting its documentation and limitations.

Tasks to be completed

1. Load the dataset, describe each feature, inspect target balance, and identify missing or implausible zero values where the dataset documentation indicates they represent missing measurements.
2. Handle missing values appropriately and split the data into training and test sets. Scale numerical features for KNN and SVM.
3. Fit a regression model to predict a continuous variable such as BMI or glucose from the remaining features; plot actual versus predicted values.
4. Train KNN and Naïve Bayes classifiers for the dataset outcome and print correct and incorrect predictions.
5. Compare Linear, Polynomial, and RBF SVM kernels.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary the number of trees and report results.
7. Run K-Means for k = 2, 3, and 4 on scaled measurements and describe the clusters without interpreting them as medical diagnoses.
8. Discuss class imbalance, dataset bias, and the difference between predictive performance and clinical usefulness.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 09

Weather Data Analysis and Rainfall Classification

Problem statement: Use historical weather observations to estimate a continuous weather variable and classify whether rainfall occurs, then group days with similar weather conditions.

Suggested dataset: Use a public weather dataset with temperature, humidity, pressure, wind, rainfall/precipitation, and a documented rain/no-rain target. Select a location and time period and cite the source.

Tasks to be completed

1. Load the weather data, parse dates, inspect missing observations, duplicates, units, and class balance.
2. Clean and transform the dataset, encode categorical fields, and scale features where appropriate.
3. Use regression to estimate a continuous variable such as maximum temperature or precipitation amount. Use a chronological test split where possible and plot actual versus predicted values.
4. Train KNN and Naïve Bayes classifiers to predict rain/no rain and display correct and incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels.
6. Train an entropy-based ID3-style decision tree and Random Forest, varying the number of trees.
7. Apply K-Means to scaled weather features for k = 2, 3, 4, and 5; interpret clusters as weather patterns.
8. Discuss why random splitting may overestimate performance for time-dependent data and identify limitations.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; elbow plot and cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 10

Wine Quality Prediction and Wine-Type Grouping

Problem statement: Analyse physicochemical measurements to predict wine quality or quality bands and discover groups of similar samples.

Suggested dataset: Use the UCI Wine Quality dataset. If using the red/white datasets together, document how the files are combined and how wine type is encoded.

Tasks to be completed

1. Load the dataset and inspect feature distributions, quality-label distribution, missing values, and duplicate records.
2. Preprocess the data and scale numerical features for KNN, SVM, and K-Means.
3. Use regression to predict the numeric quality score and plot actual versus predicted values.
4. Convert quality scores into documented categories such as lower, medium, and higher quality; train KNN and Naïve Bayes classifiers and show correct/incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary n_estimators and compare performance.
7. Run K-Means for k = 2, 3, 4, and 5; profile clusters by average feature values and quality distribution.
8. Explain class imbalance and why the model should not be presented as a definitive judgement of taste or commercial value.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; inertia/elbow plot.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 11

Human Activity Recognition from Sensor Data

Problem statement: Classify a person's recorded activity from wearable or smartphone sensor measurements and explore groups of similar observations.

Suggested dataset: Use a public Human Activity Recognition dataset, such as smartphone accelerometer/gyroscope data, or a smaller documented sensor dataset.

Tasks to be completed

1. Load the dataset, identify sensor features and activity labels, and inspect missing values and class balance.
2. Clean the data and scale numerical sensor features. Keep related records or subjects appropriately separated if subject IDs are provided.
3. Use regression to estimate one continuous sensor measurement from other sensor features and plot actual versus predicted values.
4. Train KNN and Naïve Bayes classifiers to predict activity and display correct and incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary the number of trees and compare results.
7. Apply K-Means to scaled sensor data for k = 3, 4, 5, and 6; compare clusters with activity labels for analysis.
8. Discuss generalisation across people, sensor placement, and the limitations of the dataset.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 12

Network Intrusion Detection and Traffic Grouping

Problem statement: Analyse labelled network traffic records to classify benign versus suspicious traffic and group traffic observations with similar characteristics.

Suggested dataset: Use a manageable public intrusion-detection dataset, such as a selected subset of NSL-KDD or another documented network-flow dataset. Do not scan or attack real systems.

Tasks to be completed

1. Load a manageable dataset sample and inspect features, labels, class balance, missing values, and infinite values.
2. Clean the data, encode protocol/service/state fields, and scale numerical features. Prevent leakage from identifiers or target-derived fields.
3. Use regression to predict a continuous network-flow field such as duration or bytes transferred and plot actual versus predicted values.
4. Train KNN and Naïve Bayes classifiers for benign/suspicious traffic and print correct and incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels on a manageable sample.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary n_estimators and compare performance.
7. Apply K-Means for k = 2, 3, and 4 to selected scaled traffic features and interpret the clusters.
8. Report computational constraints, class imbalance, false positives, and why results on a public dataset do not establish production security.

Evaluation measures: Regression MAE/RMSE/R^2; classification precision, recall, F1-score, confusion matrix and accuracy; cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 13

E-Commerce Customer Segmentation and Purchase Prediction

Problem statement: An online retailer wants to segment customers based on purchasing behaviour and predict whether a customer will make a purchase or respond to an offer.

Suggested dataset: Use a public online-retail/customer-purchase dataset with customer ID, order frequency, monetary value, product category, session or campaign attributes, and a documented target if available.

Tasks to be completed

1. Load and inspect the data, remove or document invalid transactions, and check missing customer/product fields and duplicate rows.
2. Aggregate transaction records into customer-level features where appropriate. Encode categorical features and scale features for KNN, SVM, and K-Means.
3. Fit a regression model to predict customer spending or order value and plot actual versus predicted values.
4. Define a clear classification target, such as purchase/no purchase or a documented response label, and train KNN and Naïve Bayes.
5. Compare SVM Linear, Polynomial, and RBF kernels and print correct and incorrect predictions.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary the number of trees.
7. Apply K-Means for k = 2, 3, 4, and 5 to customer-level numerical features and describe the customer segments.
8. Discuss privacy, leakage, and why customer segmentation should not rely on sensitive or inappropriate attributes.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score; cluster profiles and elbow plot.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 14

Air Quality Analysis and Pollution-Level Classification

Problem statement: Use air-quality measurements to estimate pollutant concentration, classify pollution levels, and identify days with similar pollution profiles.

Suggested dataset: Use a public air-quality dataset with PM2.5, PM10, NO_2, SO_2, CO, ozone, weather variables, date/time, and a documented location.

Tasks to be completed

1. Load the data and inspect measurement units, missing readings, duplicates, outliers, and date/time coverage.
2. Clean missing values appropriately, encode any categorical fields, and scale numerical features for relevant algorithms.
3. Use regression to predict one pollutant concentration from other measurements and plot actual versus predicted values.
4. Define pollution-level categories using a cited or explicitly stated threshold scheme, then train KNN and Naïve Bayes classifiers.
5. Compare SVM Linear, Polynomial, and RBF kernels and display correct and incorrect predictions.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary n_estimators and compare performance.
7. Apply K-Means for k = 2, 3, 4, and 5 to scaled pollution measurements and interpret the groups.
8. Discuss temporal patterns, missing-data limitations, and the difference between educational analysis and official health advisories.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; elbow plot and cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.

MINI PROJECT SET 15

Employee Attrition Analysis and Workforce Segmentation

Problem statement: Analyse a public employee dataset to classify historical attrition labels, estimate a continuous employment-related measure, and group employees by work-related characteristics. Use only for academic analysis, not individual employment decisions.

Suggested dataset: Use a public HR analytics/employee attrition dataset with tenure, job role, satisfaction, work-life balance, income, and attrition label. Document source and fields.

Tasks to be completed

1. Load and inspect the data, target balance, missing values, duplicates, and categorical/numerical fields.
2. Preprocess features, encode categorical columns, and scale features where appropriate. Exclude identifiers and avoid target leakage.
3. Fit a regression model to predict a continuous field such as monthly income or tenure and plot actual versus predicted values.
4. Train KNN and Naïve Bayes classifiers for the attrition label and print correct and incorrect predictions.
5. Compare SVM Linear, Polynomial, and RBF kernels.
6. Train an entropy-based ID3-style decision tree and Random Forest; vary the number of trees and compare performance.
7. Use K-Means for k = 2, 3, and 4 on selected scaled work-related features; describe group-level patterns.
8. Discuss privacy, fairness, small-sample limitations, and why model predictions must not be used to penalise or make automated decisions about employees.

Evaluation measures: Regression MAE/RMSE/R^2; classification accuracy, precision, recall, F1-score, confusion matrix; cluster profiles.

Deliverables

Submit (1) a short project report, (2) commented Python notebook/script, (3) dataset or source link, (4) plots and model-comparison table, and (5) a brief conclusion with limitations. Include sample correct and incorrect classification predictions and the Random Forest tree-count comparison.